

Date: 8/10/25

Task 10: use matplotlib module for plotting in python.

Aim: To use Matplotlib module for plotting in python.

Problem 10: I write a python programming to display a bar chart of the popularity of programming languages.

Algorithm:

1. Define two lists for programming languages and their popularity respectively
2. Find the maximum popularity value in the list.
3. Define a scaling factor to scale the bar heights within a certain limit (e.g. 50 characters)
4. For each language and popularity pair, calculate the bar height as the popularity value scaled by the scaling factor.
5. Print the chart using a loop to iterate over the programming language list:
 - a. print the language name and a separator character (e.g. "|")
 - b. use a loop to print the bar chart by printing the bar character (e.g. "*") a number of times equal to the bar height.
 - c. print the popularity value with a separator character

Program:

```
# pip install matplotlib.
```

```
import matplotlib.pyplot as plt.
```

```
languages = ["Java", "python", "PHP", "Javascript", "C#",  
             "C++"]
```

```
popularity = [22.2, 17.6, 8.8, 8, 7.7, 6.7]
```

```
plt.bar(languages, popularity, color='b')
```

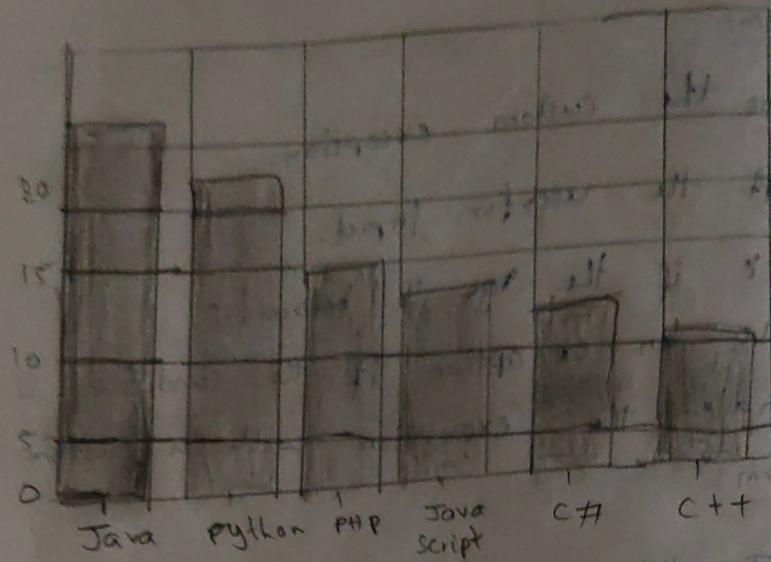
```
plt.title('popularity of programming languages')
```

```
plt.xlabel('programming languages')
```

```
plt.ylabel('popularity')
```

```
plt.show()
```

output: popularity of programming language.



programming languages.

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Problem-1: write a python programming to create a pie chart of the popularity of programming languages.

Aim: write a python programming to create a pie chart of the popularity of programming languages.

Algorithm:

1. create a list of programming languages and popularity.
2. create a pie chart using the matplotlib library.
3. set the title and legend for the pie chart.
4. show the pie chart

Program:

```
import matplotlib.pyplot as plt.
```

#step 1.

```
languages = ['Java', 'python', 'PHP', 'javascript', 'c#', 'c++']  
popularity = [22.2, 17.6, 8.8, 8, 7.7, 6.7]
```

#step 2

```
plt.pie(popularity, labels=languages, output='r.1-1p1.1.')
```

#step 3

```
plt.title('popularity of programming languages')  
plt.legend(languages, loc="best").
```

#step 4

```
plt.show().
```

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EX-NO.	10
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	
TOTAL (20)	
SIGN WITH DATE	15

Result: Thus the python program use matplotlib module for plotting is executed and verified successfully.

output:

popularity of programming language.

